# Sound recording and mixing

TP DATASIM #1 — January 29th, 2026

Instructors: Vincent Lostanlen, Morgan Buisson

We recommend you work in groups of four to six people.

## Part 1: Merlin Bird ID and Noise Capture (8h–8h45)

- Install the Merlin Bird ID app (Android/iOS). Optional: also install the Noise Capture app (Android only).

- Go outside and do some birdwatching on campus. Be extremely quiet and respectful of others. I recommend the yard between buildings A and L. Use the Merlin Bird ID to identify species. The “Sound” mode displays a spectrogram in real time. Try making mental connections between what you see on the spectrogram and what you hear on the field.

- You may also try other modes of Merlin Bird ID: “Step by Step” (easy) and “Photo” (hard).

- Try recording a birdsong on your Dictaphone app.

- If you have an Android phone and managed to install Noise Capture, measure the decibel level around you. Display spectrum, spectrogram, and map.

- When you're done, return to the TP room.

## Part 2: Recording (8h45–10h)

- Your group has to try recording at least one example of each of the following sounds. You will be evaluated on quality and critical thinking, not quantity. For each of them, write a description of the sound, giving as many details as possible while respecting anonymity: use nouns and action verbs. Add a one-sentence description of your perception of the sound: use adjectives.

a. A voice reciting a poem, in the language of your choice. Consider visiting Wikisource or other websites to find a good poem.
b. The sound of keystrokes on a computer keyboard.
c. A machine sound: heating, ventilation, printer industrial machine. If you visit the mechanical workshops of Centrale, make sure that you are allowed access and comply with safety precautions.
d. A single note from a musical instrument. If you can't find any, a whistle will do.
e. The sound of a thermal engine: car, truck, or motorcycle. If you go to a parking lot, be careful, watch around you!
f. The song or call of an animal. If you're lucky, you have already collected one in Part 1.
g. The sound of water.
h. A synthetic signal: ringtone, buzzer, phone notification, car horn. Do not trigger any emergency alarm!

- When you're done, return to the TP room.

# Part 3: Mixing (10h–10h45)

- Parts 3a and 3b may be conducted in parallel by splitting your group into two subgroups. Both rely on Audacity.

## Part 3a. Mixing voice and music

- Explore Audacity's documentation
- Download a copyright-free music, for example on Jamendo. Choose it so that it suits the poem you recorded.
- Create two tracks using Audacity, allocate one for the voice and another for the music. Use gain, amplification, and EQ to mix the two tracks appropriately.

## Part 3b. Mixing environmental sounds

- Explore Audacity's documentation
- Import your recordings
    - Open Audacity and import several sounds you have recorded in Part 2.
    - Each sound should appear on a  separate track.
- Crop regions of interest
    - Listen to each recording.

    - Using the selection tool, select a time interval that best represents the sound event you want to study.
    - Crop the tack to this region.
    - The goal is not to isolate a meaningful acoustic event.
- Equalize volume across samples
    - Adjust the loudness of each sound (Effect/Normalize or Effect/Amplify).
    - Apply the same target level to all tracks (for example, peak at -1dB).
    - Listen again and verify that differences in loudness are not due to the recording conditions anymore.
- Visualize the spectogram
    - For each track, click on the track menu and select Spectrogram.
    - Adjust settings (window size, frequency scale) if needed.
    - Observe how different sound categories produce distinct time-frequency patterns.
    - Try to link what you hear to what you see.

# Part 4: Collective listening (10h45–11h30)

Everyone meets in the TP room. You will be invited to playback your two mixes (4a and 4b) to the other groups. ~ 8 minutes per group including Q&A.

# Part 5: FreeSound (11h30–noon)

- Create a FreeSound account (one per group)
- Prepare your sound files
    - Export your processed sounds from Audacity.
    - Use clear, consistent filenames.
    - Make sure each sound respects anonymity and does not contain private or sensitive information.
- Upload sounds
    - For each sound you upload, fill-in the required information (title, description, tags).
- Choose a license
    - Select an appropriate Creative Common License.
    - Verify what the license allows others to do with your sound.